

Executive Summary – Employee Salary Analysis

1. Project Overview

The Employee Salary Analysis project provides an overview of average annual salaries across different job titles, departments, ethnicities, business units, gender groups, and hiring years. The dashboard uses interactive visualizations and filters to help understand salary patterns and differences across various employee categories.

The analysis covers employees hired across the displayed period of 2016–2019 and provides a comparative view of average salary across organizational segments.

2. Key Insights

- **Average Salary by Job Title**

The job-title analysis shows considerable variation in average annual salary across different positions. Senior and leadership positions such as Vice President, Director, and Manager-level roles appear toward the higher end of the salary distribution, while several analyst and specialist positions have comparatively lower average salaries.

This indicates that job responsibility and seniority are important factors associated with differences in average salary.

- **Average Salary by Department**

The department-level treemap shows differences in average salary across departments including:

- Engineering
- Finance
- Accounting
- Human Resources
- IT
- Marketing
- Sales

The visualization indicates that salary levels vary across departments, with Finance and Engineering occupying relatively prominent positions in the displayed analysis.

- **Average Salary by Ethnicity**

The dashboard shows the following average annual salaries:

Ethnicity Average Salary

Asian 1,17,628

Latino 1,11,536

Caucasian 1,09,345

Black 1,09,022

The displayed data shows differences in average salary across the four groups. These figures are descriptive averages from the dataset and do not by themselves explain the reasons for the differences.

- **Average Salary by Business Unit**

The dashboard compares four business units:

Business Unit Average Salary

Corporate 1,20,776

Specialty Products 1,16,796

Research & Development 1,12,490

Manufacturing 1,03,651

Corporate has the highest displayed average salary, while Manufacturing has the lowest among the four business units.

This suggests that organizational function and business-unit structure may be associated with differences in average compensation.

- **Gender and Hire Date Analysis**

The dashboard includes an interactive Gender filter with Female and Male options, allowing salary analysis to be examined separately by gender.

A Hire Date filter is also provided for year-wise analysis from 2016 to 2019, allowing users to investigate salary patterns across different hiring periods.

3. Overall Business Insights

The analysis highlights several important patterns:

1. Average salary varies considerably across job titles and organizational roles.
2. Senior and leadership positions appear toward the higher end of the job-title salary distribution.

3. Corporate has the highest displayed average salary among the four business units at 1,20,776.
4. Manufacturing has the lowest displayed business-unit average at 1,03,651.
5. Among the displayed ethnicity categories, the highest average is 1,17,628 for Asian employees, while the lowest is 1,09,022 for Black employees.
6. Department-level analysis shows variation in average salary across Finance, Engineering, Accounting, HR, IT, Marketing, and Sales.
7. The Gender and Hire Date slicers make it possible to conduct more detailed comparisons across employee groups and hiring years.

4. Recommendations

1. Review Salary Differences by Job Level

Conduct regular salary benchmarking across similar job titles and responsibility levels to ensure that compensation structures remain consistent and competitive.

2. Analyze Department-Level Compensation

Departments with substantially different salary levels should be reviewed based on factors such as role complexity, experience requirements, skills, and market compensation.

3. Review Business Unit Salary Structures

The difference between the average salaries of Corporate (1,20,776) and Manufacturing (1,03,651) suggests an opportunity to review compensation structures across business units.

4. Use Hiring-Year Analysis

Compare salaries across the 2016–2019 hiring years to identify changes in compensation levels and understand how salary structures have evolved over time.

Conclusion

The Employee Salary Analysis dashboard provides a consolidated view of compensation patterns across different organizational dimensions. The analysis demonstrates that average salary varies by job title, department, business unit, and employee demographic category.

The dashboard can support HR and management teams in salary benchmarking, compensation planning, workforce analysis, and monitoring of compensation equity. Combining the overall averages with factors such as job level, experience, tenure, and hiring year would provide a more meaningful basis for understanding the observed salary differences.